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MILITARY SPECIFICATION

MODEM, RADIO TELETYPEWRITER SET MD-522( )/GRC

1. SCOPE

1.1 This specification covers Teletypewriter Set MD-522( )/GRC which is designed to provide a radio terminal capable of simultaneously transmitting and receiving narrow shift teletype (85 Hz), wide shift teletype (850 Hz), frequency diversity teletype (85 Hz), simultaneous voice and narrow shift teletype (85 Hz), or voice only.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	Pallet, Materials-handling, Wood(General Construction Requirements)
QQ-S-781	Strapping, Steel, Flat and Seals
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated Plywood
PPP-B-621	Box, Wood, Nailed and Lock Corner
PPP-B-636	Box, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated Triple-Wall
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-S-760	Strapping, Nonmetallic (and Connectors)
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water-Resistant (for Carton Sealing)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

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SPECIFICATIONS (Cont)

MILITARY

MIL-P-116	Preservation, Methods of
MIL-T-152	Treatment, Moisture And Fungus-resistant, Of Communications, Electronic And Associated Electrical Equipment
MIL-V-173	Varnish, Moisture And Fungus Resistant (For Treatment Of Communications, Electronic and Associated Equipment)
MIL-I-6869	Impregnants for Aluminum Alloy and Magnesium Alloy Castings
MIL-P-11268	Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	Marking Of Electronic Items
MIL-P-13949	Plastic Sheet, Laminated, Copper-Clad (For Printed Wiring)
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-S-19500	Semiconductor Devices, General Specification For
MIL-I-46058	Insulating Compound, Electrical (for Coating Printed Circuit Assemblies)
MIL-P-55110	Printed Wiring Board

STANDARDS

FEDERAL

FED-STD-595	Color

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized Unit Loads 40 in. x 48 in. 4-way Partial and 4-way Pallets
MIL-STD-252	Wired Equipment Classification of Visual and Mechanical Defects
MIL-STD-275	Printed Wiring for Electronic Equipment
MIL-STD-276	Impregnation of Porous Non-ferrous Metal Castings
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of

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STANDARDS (Cont)

MILITARY

MIL-STD-750	Test Methods for Semi-conductor Device
MIL-STD-781	Reliability Test, Exponential Distribution
MIL-STD-810	Environmental Test Methods
MS-3114	Connectors Receptacle, Electronic, Solder, Type, Box Mounting, Flange, No. 4 Holes, Bayonet Coupling

DRAWINGS

ELECTRONICS COMMAND

SC-A-46600	Preproduction Sample Approval in Lieu of Qualification Requirements in Specifications for Semiconductor Devices and Electron Tubes
SC-B-61578	Grounding to Chassis
SC-D-52115	Cable Assembly Special Purpose, Electrical CX-10071( )/U(10'0")
SC-DL-502723	Receiver, Case Assembly
SC-DL-502745	Cross Bar Assembly(s)
SC-DL-502747	Mounting MT-3140( )/GRC-106
SC-DL-583300	Modem, Radio Teletypewriter Set MD-522( )/GRC
SM-A-583645	MTP for Transmitter Module
SM-D-145895	Frequency Diversity Test Fixture
SM-D-583656	Error Rate Test Set
SM-D-583659	Electrical Schematic, System Test Fixture

(Copies of specifications, Standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies).

3. REQUIREMENTS

3.1 Description - Modem, Radio Teletypewriter Set MD-522( )/GRC is specifically designed to perform all necessary modulation and demodulation functions when used with a special family of single sideband, high frequency radio teletypewriter sets. Representative sets in this group are:

Radio Teletypewriter Set AN/GRC-122( )
Radio Teletypewriter Set AN/GRC-142( )
Radio Teletypewriter Set AN/VSC-2( )
Radio Teletypewriter Set AN/VSC-3( )

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Modem MD-522( )/GRC operation is not limited to these radio sets, however, and can be used with other equipments having compatible input and output characteristics. Modem MD-522( )/GRC will convert standard Baudot Teletype code to audio tones for modulating radio transmitters and convert received audio tones to Baudot code signals. The following modes of operation (transmit and receive) are made available by means of a front panel switch.

a.	Teletype 850 Hz frequency shift	Space freq. 2425 Hz Mark freq. 1575 Hz
b.	Teltype 85 Hz frequency shift	Space freq. 2847.5 Hz Mark freq. 2762.5 Hz
c.	Voice only	
d.	Teletype 85 Hz frequency shift diversity	Space freq. 2847.5 Hz and 467.5 Hz Mark freq. 2762.5 Hz and 382.5 Hz
e.	Teletype 85 Hz frequency shift plus voice	Space freq. 2847.5 Hz Mark freq. 2762.5 Hz

Either one way reversible or full duplex operation is possible as the transmit and receive functions of the equipment may be operated independently.

3.2 First Article Samples.- The contractor shall furnish First Article Samples of the unit for approval, if required by the invitation for bids and contract (see 4.3 and 6.2(d) (1)).

3.3 Construction . - The equipment shall be constructed in accordance with the requirements of this specification and of Drawing SC-DL-583300 (including the requirements for parts, materials and processes thereon).

3.4 Parts, Materials and Processes: General. - In addition to the requirements of this specification, the requirements of Specification MIL-P-11268, including the selection requirements therein, shall apply.

3.4.1 Finish, Protective (see 4.4). - Equipment shall be given protective finish in accordance with MIL-F-14072. This includes finish of hardware type items, such as handles, hinges, screws, etc. and necessary touch-up after mounting. Unless otherwise specified, the final paint film on type I surfaces (see MIL-F-14072), exposed to view during operation, shall be a smooth, semi-gloss (olive drab) green, in color conforming to No. X24087 of FED-STD-595. When some other color or gloss is referred to in the equipment specification, but the details are not specified, the contractor shall apply to the contracting officer for information regarding the required finish. For interference reduction purposes, interface surfaces of structures and bonding connections between cases and mounting bases shall have a finish for electrical contact.

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3.4.2 Marking (see 4.4). -

3.4.2.1 Conformance. - Marking shall conform to MIL-M-13231. Front panel marking shall be Group I as described in MIL-M-13231.

3.4.2.2 Visibility of Markings. - Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.4.2.3 Serial Numbers. - Modem, Radio Teletypewriter Set MD-522( )/GRC shall be consecutively serial numbered. Mounting MT-3140( )/GRC-106, if included in the invitation for bids and contract, shall also be consecutively serial numbered.

3.4.3 Cleaning. -

3.4.3.1 Parts. - After fabrication, parts shall be cleaned in accordance with requirement 9 of MIL-STD-454, or as specified in an applicable document.

3.4.3.2 Units. - After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, such cleaning shall also be performed before final assembly of the equipment.

3.4.4 Grounding to chassis. - Grounding to chassis shall conform to drawing SC-B-61578.

3.4.5 Tropicalization of material and parts (see 4.7.9). - Fungus inert materials shall be in accordance with requirement 4 of MIL-STD-454.

3.4.5.1 Overall treatment (moisture and fungus proofing treatment(MFP)). - When overall tropicalization treatment of equipment is a contractual requirement, a moisture and fungus resistant coating of varnish conforming to MIL-V-173 shall be applied in accordance with MIL-T-152. The moisture-fungus resistant coating shall be applied after the equipment has been cleaned to remove such contaminants as corrosion products, dust, fingerprints, lubricating oils, mold release agents, sand, and solder fluxes. When tape is used on parts of equipment that are masked against treatments, the tape shall be an electrical grade. In the application of the MFP coating, care shall be taken to assure that the coating does not affect circuit constants and thereby degrade performance.

3.4.5.2 Fungus (see 4.7.9). - The equipment (including accessories) shall show no evidence of visible fungus or corrosion when subjected to the test specified in 4.7.9. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

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3.4.6 Integrated circuits (IC's) (see 4.7.10). - The electrical parameters of IC's shall be those that will enable the equipment to meet its performance and other requirements, and also those that will establish the stability and reliability of the IC irrespective of the application. After subjection to the tests specified in 4.7.10, the IC's shall meet these electrical parameters and visual and mechanical inspection as applicable. IC's shall be legibly marked with the manufacturer's part number.

3.4.7 Printed wiring assemblies.

3.4.7.1 Printed wiring boards. - Printed wiring boards used in the construction of printed wiring assemblies, shall be a product which has passed the requirements of Specification MIL-P-55110, Standard MIL-STD-275, master drawing and master pattern. (NOTE: The printed wiring boards shall meet all the requirements of the latest issue of MIL-P-55110 and MIL-STD-275, except that the size and shape of the board, the size and location of all holes therein, and the shape or arrangement of both conductor and non-conductor pattern or elements, shall be per the master drawing and the master pattern).

3.4.7.2 Laminate. - The laminate shall be type BG or GH in accordance with Specification MIL-P-13949, unless otherwise specified in the master drawing.

3.4.7.3 Plating. - Plating of printed wiring boards shall be in accordance with Specification MIL-P-55110 and the master drawing.

3.4.7.4 Protective coating. - Printed wiring assemblies shall be protected on the printed wiring side of the boards with an epoxy-type conformal coating to protect the pattern against moisture, dust and other contaminants. The coating shall be Class A, Type ER or Class A, Type PUR in accordance with Specification MIL-I-46058. For areas where protective coating cannot be applied without impairment of electrical or physical parameters, the contractor shall coordinate with the Contracting Officer's Technical Representative, etc. for procedures for masking parts before applying the coating.

3.4.7.5 Solderability. - Solderability of printed wiring boards shall conform to Specification MIL-P-55110.

3.4.8 Semiconductor devices. - Unless otherwise specified in equipment drawings, all semiconductor devices shall conform to Requirement 30 of MIL-STD-454. Testing shall be in accordance with MIL-STD-750.

3.5 Service Conditions (see 4.7). - The equipment shall meet the following service conditions; where a test is referenced, meeting the test shall be considered as compliance with the requirement.

3.5.1 Temperature (see 4.7.1). -

3.5.1.1 Low. - During testing as specified in 4.7.1.1, the equipment shall meet the following requirements:

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Paragraph	Measurement	Performance
3.7.1, 3.7.3, 3.7.7, 3.7.12, 3.7.13, 3.7.21	System Test	4.8.2
3.7.9	BFO Stability	4.8.11

After testing, the equipment shall meet full specification performance for these measurements.

3.5.1.2 High. - During testing as specified in 4.7.1.2, the equipment shall meet the following requirements:

Paragraph	Measurement	Performance
3.7.1, 3.7.3, 3.7.7, 3.7.12, 3.7.13, 3.7.21	System Test	4.8.2
3.7.9	BFO Stability	4.8.11

After testing, the equipment shall meet full specification performance for these measurements.

3.5.2 Humidity (Moisture-resistance) (see 4.7.2). - During testing as specified in 4.7.2, the equipment shall meet the following requirements:

Paragraph	Measurement	Performance
3.7.1, 3.7.3, 3.7.7, 3.7.12, 3.7.13, 3.7.21	System Test	4.8.2
3.7.9	BFO Stability	4.8.11

After testing, the equipment shall meet full specification performance for these measurements and there shall be no mechanical deterioration as evidenced by corrosion of metal parts or binding of rotating parts.

3.5.3 Altitude (see 4.7.3). - The equipment shall meet full specification requirements for the following measurements both during and after testing as specified in 4.7.3.

Paragraph	Measurement	Performance
3.7.1, 3.7.3, 3.7.7, 3.7.12, 3.7.13, 3.7.21	System Test	4.8.2

3.5.4 Vibration, Internal (see 4.7.4). - With the equipment mounted on the vibration table, as specified in 4.7.4, the amplitude of vibration of any part, subassembly, module or structual member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any

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frequency between 10 and 55 hertz.

3.5.5 Vibration, Equipment (see 4.7.5). - The equipment shall be vibrated as specified in 4.7.5. The amplitude of vibration of the mounted equipment shall not exceed two times the amplitude of the applied vibration at any frequency between 10 and 55 hertz. The equipment shall be operated continuously and shall meet the following requirements before, during and after the vibration test:

Paragraph	Measurement	Performance
3.7.1, 3.7.3, 3.7.7, 3.7.12, 3.7.13, 3.7.21*	System Test	4.8.2

*During vibration, error rate measurements shall be made using a different mode for each plane.

3.5.6 Bounce and Shock. - The equipment shall be capable of meeting the requirements of Table 1.

Table 1 - Bounce and Shock

Paragraph Heading	Test Paragraph	Performance
Bounce	4.7.6	Specified performance (Note A) No physical damage
Shock Test: Bench-handling	4.7.7	Specified performance (Note B) No physical damage

NOTE A - Minor paint abrasion is permissible. During and after testing as specified in 4.7.6, the equipment shall meet the following requirements:

Paragraph	Measurement	Performance
3.7.1, 3.7.3, 3.7.7, 3.7.12, 3.7.13, 3.7.21	System Test	4.8.2

NOTE B - After testing as specified in 4.7.7, the equipment shall meet the following requirements:

Paragraph	Measurement	Performance
3.7.1, 3.7.3, 3.7.7, 3.7.12, 3.7.13, 3.7.21	System Test	4.8.2

3.5.7 Immersion (see 4.7.8). - The equipment shall show no evidence of leakage after it is tested as specified in 4.7.8.

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3.6 Electromagnetic Compatibility (see 4.4). - The equipment shall meet the electromagnetic compatibility requirements of MIL-STD-461 (Table A-1-MIL-E-55301) and MIL-STD-462 for Class 1 equipments.

3.7 Electrical requirements (see 4.8).

3.7.1 FSK Modes (see 4.8.2). - The teletype modem unit shall be capable of duplex sending and receiving the following forms of teletype and/or voice information obtained as audio across Pins J and D (Pins C and D for voice only) of the Receive/Transmit Audio Connector J2 for transmitting and applied as audio across Pins A and B of the Auxiliary Receiver Audio Connector J1 for receiving.

a. Conventional 850 Hz shift with a SPACE frequency of 2,425 ± 25 Hz and a MARK frequency of 1,575 ± 25 Hz.

b. Voice plus 85 Hz shift with a SPACE frequency of 2,847.5 ± 2 Hz and a MARK frequency of 2,762.5 ± 2 Hz.

c. Frequency Diversity 85 Hz shift, with SPACE frequencies 2,847.5 ± 2 Hz and 467.5 ± 2 Hz and MARK frequencies of 2,762.5 ± 2 Hz and 382.5 Hz ± 2 Hz.

d. Voice only (full available bandwidth).

e. 85 Hz shift with a SPACE frequency of 2,847.5 ± 2 Hz and a MARK frequency of 2,762.5 ± 2 Hz.

3.7.2 Cross talk (see 4.8.3). - In the Voice plus 85 Hz shift mode of operation (3.7.1(b)), the level of crosstalk in either the Teletype or Voice channels shall be at least 46 db below the desired signal level.

3.7.3 Teletype tone stability (see 4.8.5). - The stability of the teletype MARK and SPACE frequencies shall be sufficiently high so that during a six month period with variations in ambient temperature as specified herein, the frequency of the audio tone, without adjustment, shall be within the limits of 3.7.1.

3.7.4 Teletype tone frequency adjustment (see 4.8.6). - Internal adjustments shall be provided to allow the four audio tone frequencies to be adjusted to within 0.1 Hz of the nominal frequency. The other two frequencies (diversity tones 467.5 Hz and 382.5 Hz) shall be adjustable to within 0.2 Hz of the nominal frequency.

3.7.5 Transmit tone reversal (see 4.8.7). - A switch on the transmit module shall reverse the MARK and SPACE tones. This switch shall be labeled "Normal" and "Reverse". In the "Normal"position, the MARK frequency shall be the lower frequency.

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3.7.6 Receive polarity reversal (see 4.8.7). - A switch on the front panel shall reverse the polarity of MARK and SPACE signals when the modem is used in the receive mode. In the reverse position, MARK will be a no current condition and SPACE will provide current from the modem. The switch position shall be labeled "Normal" and "Reverse".

3.7.7 Mark and Space tone levels (see 4.8.8). - The audio output level of all transmitted mark and space tones in any mode of operation shall be within 1.0 db of each other. This relative level shall be maintained within 2.0 db over the operating temperature range -40°F to ± 150°F of the equipment.

3.7.8 Current regulated supplies (see 4.8.9). - The two loop supplies within the modem shall be current regulated with a choice of either 20 ma or 60 ma output, provided by a switch on the front panel. The chosen current output shall remain constant within plus or minus 3 milliamperes over a load variation of from 0 to 1,300-ohms in series with a loop inductance equivalent to three TT-98 teletype machine selector magnets, a simulated mile of wet WD-1/TT field wire and 50 ohms resistance. The supplies shall not be damaged if operated without a load or if the loop is shorted.

3.7.9 BFO (see 4.8.10). - A BFO in the receive circuit, after a five minute warm-up, shall permit adjustment of the teletype tones in the receiver discriminator to compensate for frequency drift of the received signal. The stability of BFO shall be such that its contribution to the tone "frequency drift" shall not be more than ± 25 Hertz under any of the combined service conditions of 4.7. The range of the BFO shall be at least plus or minus 1000 Hertz. This control shall function only when using the 850 Hertz FSK mode; however, the oven shall be on in all operate modes.

3.7.10 Tone vs Voice level (see 4.8.12). - When using 85 Hertz shift teletype plus voice mode, an internal adjustment shall be provided to vary the tone levels of the teletype channel and the 50 ohm voice channel level to provide for optimum relationship between voice and teletype power in the transmitted signal.

3.7.11 Audio levels and distortion (see 4.8.12). - The audio level of the transmitted teletype tones appearing between pin J and pin D of the audio connector labeled "Rec Trans Audio" shall be a nominal 200 millivolts across 600 ohms. The audio output to a headset shall be at least 10 milliwatts across 600 ohms and shall be on pin A and ground of the audio connectors labeled "Microphone" and "Speaker and Remote". The loudspeaker output between pin L and ground on the "Speaker and Remote" shall be at least 2 watts across 600 ohms. The maximum distortion at the 10 milliwatt output and at the 2 watt output shall be not more than 5%.

3.7.12 Teletype keying capability (see 4.8.2). - The teletype modem shall operate at 60 to 100 words per minute using the standard Baudot Teletype Code.

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3.7.13 Teletype distortion (see 4.8.2). - The average distortion (mark or space bias) shall not be greater than 5% for any mode of 3.7.1 and the instanteous distortion (jitter) of the teletype transitions shall not exceed 500 microseconds for 85 Hz shift and 1.3 milliseconds for 850 Hz shift at any speed from 60 to 100 words per minute and using the DC loop loads specified in 3.7.8.

3.7.14 Teletype modem configuration. - The unit shall be completely transistorized except for cathode ray tube and no mechanical keying devices shall be permitted.

3.7.15 Receptacles. - Front panel receptacles shall be of the "PT" type except for the audio 10 pin connectors marked "Microphone"and "Speaker-Remote" and the power connector which shall be as specified. Each "PT" type receptacle shall have a distinguishing character to preclude improper connection of cables. The "DX Send/OWR" receptacle shall be type MS-3114E-12-3S, the "DX-RCV/Pony" receptacle shall be type MS-3114E-12-3SY, the "Auxiliary receptable" shall be type MS-3114E-12-10SY, the "Aux. Rec. Audio" receptacle shall be type MS-3114E-12-10P, the "Rec-Trans Audio" receptacle shall be type MS-3114E-12-10S. The power connector shall be type UW2020MA00. All receptacles except the power receptacle shall be supplied with captive covers.

3.7.16 Scope (see 4.8.13). - A cathode ray scope shall be provided to facilitate tuning, operation and maintenance. Internal adjustments shall be provided, accessible from the top of the scope module as mounted in the modem, for the adjustment of all of the normal scope functions (i.e., horizontal and vertical positioning and amplifier gain, focus, etc.) except intensity. The intensity control shall be on the front panel. The scope display shall be biased off when no signal is present, and in the "Voice only" mode. The intensity of the scope pattern shall be sufficient to permit its observation on a sunny day with the scope face shaded.

3.7.17 Test Meter (see 4.8.24). - When tested as specified in 4.8.24, the test meter shall give an accurate indication of the appropriate meter function as controlled by the meter switch. The meter indications shall have an accuracy of at least 90% referred to the nominal value for the switch setting.

3.7.18 Frequency diversity (see 4.8.4). - The diversity combining network shall combine the output of the two discriminators with no degradation in signal. Whenever the two inputs differ in level by 3 $\pm$ 1 db or more the channel with the better signal shall be selected. The choice between channels shall be based on signal-to-noise ratio. The scope pattern in the diversity mode shall be switched so that it operates always on the channel being used.

3.7.19 Prime power (see 4.8.18).- The teletype modem shall operate from an external DC voltage source of 22 to 30 VDC. Reduced performance is permitted between 22 and 24 VDC. The input shall employ a circuit breaker to protect

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the equipment. The circuit breaker shall serve as the "On-Off" control for the modem. When tested as specified in 4.8.19 the circuit breaker (trip point) shall open between 31.3 to 34.7 VDC.

3.7.19.1 Transient protection (see 4.8.14). - The equipment shall be protected against voltage transients which cause the input voltage on the input power line to go as high as 100V peak at any duration and repetition rate. The equipment should continue to operate properly on 100V transients of 50 MS or less at a repetition rate of 2 per second or less.

3.7.19.2 Reverse Current (see 4.8.14). - The modem shall be protected against damage in the event the polarity of the prime power DC voltage is reversed.

3.7.20 Audio response (see 4.8.12). - In the "Voice only" mode, the audio response of the modem shall not degrade the input signal in transmit operation. On receive operation, "Voice only" mode, the Audio Response shall not degrade more than 1 db total from 300 to 3000 Hertz and 2 db total from 3000 to 3500 Hertz. The transmitted and received voice signal in the "85 Hz plus voice" mode shall conform to the following:

200 and 2260 Hz not more than -3 db referenced to 1000 Hz

2500 Hz at least -30 db referenced to 1000 Hz

3.7.21 Error rate vs signal-to-noise ratio (see 4.8.2). - The error rate of the modem when operated as a system (back-to-back) and modulated by reversals at 100 words per minute, 85 Hz or 850 Hz shift, shall fall within the shaded areas of Figures 1 and 2 "Typical model error rate performance".

3.7.22 Automatic mark hold (see 4.8.15). - In the absence of a received teletype signal, the modem shall revert to a condition simulating "Mark Hold" (i.e., there shall be a steady current in the teletype loop). This condition shall occur even though there is a noise input to the modem. In no case shall the Automatic Mark Hold function occur with signal-to-noise ratios of -9 db for 85 Hz mode and -6 db for 850 Hz mode or better. The front panel switch may be used to disable the Automatic Mark Hold circuit.

3.7.23 Audio Squelch (see 4.8.16). - The modem voice squelch circuit shall squelch the audio output to headset and loudspeaker in the absence of a voice signal. The squelch shall not operate to squelch the audio output with signal-to-noise ratios of plus 6 db or better. A squelch sensitivity control shall be provided on the front panel.

3.7.24 Front panel controls. - The front panel controls of 3.7.6, 3.7.8, 3.7.16, 3.7.22 and 3.7.23 shall be located behind a hinged access door on the front panel. The access door shall be held closed by a screw fastener which can be turned by the operator without the use of a tool. The door shall be hinged downward and shall not cover or conceal any other front panel controls or indicators when open.

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3.7.25 One-way duplex (see 4.8.21). - The modem shall operate both in the duplex mode and the one-way reversible mode. When operating duplex, each of the two loop battery supplies shall be used; one in the receive loop to the "DX RCV/PONY" receptacle, and one in the transmit loop to the "DX SEND/OWR" receptacle. In the one-way reversible mode, only one loop battery supply shall be used for both the transmit loop and the receive loop. It shall appear at the "DX SEND/OWR" receptacle. The loop battery supply in this mode shall be switched between the two teletype loops by means of the Send-Receive switch. In the one-way reversible mode, the other loop battery supply shall be independently connected to the "DX RCV/PONY" receptacle to be used to power a teletype "Pony" circuit, This function shall be accomplished by means of a front panel switch labeled "One Way" "Duplex". This function shall also appear on the "Auxiliary" receptacle on the front panel as a closure to ground on pin A in duplex.

3.7.26 Send-receive switch (see 4.8.22). - A "Send-Receive" switch shall be located on the front panel of the modem and shall provide the necessary switching function within the modem to accomplish the "one-way reversible" switching requirements of 3.7.25 and any other functions that are required in one-way operation. The "one-way reversible" switch function shall also be available as a closure to ground in "Send" on pin D of the "Auxiliary" receptacle on the front panel. When the "Send-Receive" switch is in the "Receive"position, remote control of this function shall be accomplished by grounding pin E in the "Auxiliary" receptacle. Grounding this pin shall change the modem from the "Receive" to the "Send" condition.

3.7.27 Audio functions (see 4.8.23). - The receive audio amplifier circuitry in addition to operating in the "Voice only" mode and the "85 Hz plus Voice" mode, shall provide an audio output of the received teletype tone in the teletype only modes. In the 85 Hz plus voice mode, the only audio output shall be the voice portion of the received signal. The switching function for the selection of the proper audio output shall be controlled by the mode selector switch. In the 85 Hz plus voice and the "Voice" transmit modes, both a 50 ohm and 600 ohm voice channel will be provided. The 600 ohm channel shall provide a minimum transmit output level of 200 mv for a 200 mv input signal.

3.7.28 720 Hour run (see 4.8.20). - When tested as specified in 4.8.20, a failure occurring within the first 72 hours, or if more than 3 failures occur during the remainder of the test, this will be cause for rejection. Also if the allowed failures are repeats from the same cause, the equipment shall be rejected. The daily tests and the tests at the end of the run shall meet specified requirements.

3.7.29 Ground post. - The ground post shall afford a good grip for connection to Nr. 10 American Wire Gauge stranded wire.

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3.8 Case. - The case used to house the modem shall be constructed in accordance with Drawing SC-DL-502723.

3.9 Ancillary equipment. - If procured by the contract, the following items shall meet the applicable requirements of workmanship inspection and the service conditions tests (see 6.2(c)).

3.9.1 Mounting MT-3140( )/GRC-106. - The mount assembly shall be constructed in accordance with Drawing SC-DL-502747.

3.9.2 Cross bar assembly. - The cross bar assembly shall be constructed in accordance with SC-DL-502745.

3.9.3 Cable assembly, special purpose, electrical CX-10071( )/U (10' 0"). - The cable assembly shall be constructed in accordance with Drawing SC-D-52115.

3.9.4 Impregnation of aluminum castings. - All exterior aluminum castings shall be impregnated in accordance with Standard MIL-STD-276, using type II impregnating material in accordance with Specification MIL-I-6869. Castings which have been impregnated shall be marked "IMP". The impregnation shall be used to correct general seepage only and shall not be used to correct poor foundry techniques or significant porosity. (see 4.4).

3.10 Interchangeability (see 4.9). - Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454 and shall be physically and functionally interchangeable, without modification of such items or of the equipment.

3.11 Preconditioning (see 4.6). - All equipments, including First Article Sample, produced under this specification shall be preconditioned in accordance with the following:

3.11.1 Bounce preconditioning. - The equipment shall be capable of meeting the requirements specified herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6.1 (also see 4.5).

3.11.2 Burn-In/Debugging. - Each equipment on order shall be subjected to the system test and shall meet the system test requirements after the burn-in/debugging of 4.6.2.

3.12 Air seal. - When tested as specified in 4.10, the decrease in vacuum shall not exceed 0.01 pounds per square inch, gauge (psig), per minute.

3.13 Air seal test, provisions for. - The equipment enclosures shall provide means for enabling performance of the air seal test (see 4.10).

3.14 Technical literature, tools, and running spare parts. - Technical literature, tools, and running spare parts shall be furnished if specified in

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the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order (see 6.2(g) and (h)).

3.15 Workmanship. - The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

a. In MIL-P-11268:

General requirements for plastic material and plastic parts.

Wiring and cabling, including:

Slack
Protection
Splicing
Clearance
Connection, general
Grounding, general
Shielding on wire and cable
Flux and cleaning agent for soldering
General requirements for securing of parts
Cleaning prior to welding
Cleaning after welding
Soldering and brazing
Riveting
Process for welding
Mechanical operation of controls
Gears
Fastener, hardware
Microelectronics/integrated electronics and modular design
Printed wiring and encapsulated assemblies

b. In this specification:

3.3	Construction
3.4.1	Finish, protective
3.4.2	Marking of equipment
3.4.2.3	Serial numbers
3.4.3	Cleaning
3.4.4	Chassis, grounding
3.4.5	Tropicalization of material and parts
3.4.6	Integrated circuits
3.4.7	Printed wiring assemblies
3.4.8	Semiconductor devices
3.7.24	Front panel controls
3.7.29	Ground post
3.8	Case
3.9	Ancillary equipment

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Unless otherwise specified, test equipment for Government verification inspection shall be made available by the contractor at no additional cost to the Government.

4.2 Classification of inspection. - Inspection shall be classified as follows:

a. First Article inspection (does not include preparation for delivery) (see 4.3).

b. Inspection covered by subsidiary documents (see 4.4).

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.13).

4.3 First Article. - Unless otherwise specified in the contract or purchase order, the First Article inspection shall be performed by the contractor.

4.3.1 First Article units. - The contractor shall furnish four (4) First Article units of Modem, Radio Teletypewriter Set MD-522( )/GRC.

4.3.2 First Article inspection. - The First Article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, preconditioning specified in 4.6 and the inspections specified for Group A, Group B and Group C (see Table III, Table IV and Table V, respectively). The inspections shall be performed in the following order: 4.4, 4.6, Group A and Group B for all units, and designated units for specific Group C tests in the order shown in Table II with the balance of Group C tests performed on all First Article units.

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TABLE II

Inspection(Note 1)	Unit 1	Unit 2	Unit 3	Unit 4
Vibration, internal		1		
Vibration, equipment	1			
Temperature		2		
Bounce	2			
Shock, bench handling	3			
Humidity	4			
Fungus (Note 2)			1	
Immersion		3		
720 Hour				1

Note 1: Other First Article inspection may precede, follow, or be interspersed between the above tests.

Note 2: The equipment shall be thoroughly washed, cleaned, dried and refurbished, if necessary, before proceeding with subsequent tests.

4.4 Inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification:

Inspections	Req Para
Finish, Protective	3.4.1
Marking of equipment	3.4.2
Integrated circuits	3.4.6
Printed wiring assemblies	3.4.7
Semiconductor devices	3.4.8
Casting impregnation	3.9.4
*Electromagnetic compatibility	3.6
Workmanship	3.15

*This test shall be performed during First Article inspection only.

4.5 Quality conformance inspection of equipment before preparation for delivery (see 3.11 and 4.6).- Each unit of each lot of equipment shall be preconditioned prior to inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement as a result of preconditioning shall be classified as a defect and recorded as such. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government, at its discretion, may perform all or any part of the specified

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inspection, to verify the contractor's compliance with specified requirements (see 6.8). Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A Inspection. - Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table III. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection utilizing the general inspection levels and AQL's indicated in Table III.

Table III - Group A Inspection

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Visual and Mechanical	3.15	4.11		
Modules				
Front Panel and Chassis			0.65%	2.5%
Receiver Audio and BFO			0.65%	2.5%
Scope			0.65%	2.5%
Transmitter			0.65%	2.5%
Receiver			0.65%	2.5%
Loop Battery			0.65%	2.5%
Printed circuit board assemblies			0.65%	2.5%
Assembled Modem			1.0%	4.0%
Case			1.0%	4.0%
Running Spare Parts **			1.0%	4.0%
Electrical ***				
System Test	3.7.1 3.7.3 3.7.7 3.7.12 3.7.13 3.7.21	4.8.2	1.5 dphu * for the group	
Teletype tone frequency adjust	3.7.4	4.8.6	1.5 dphu* for the group	
BFO range	3.7.9	4.8.10	1.5 dphu* for the group	
Front panel controls and receptacles	3.7.24 3.7.25 3.7.26 3.7.27	4.8.17 4.8.21 4.8.22 4.8.23 4.8.24	1.5 dphu* for the group	
Air seal	3.12	4.10	1.0%	
Running spare parts	3.14	(See Note)	1.5%	

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*Defects per hundred units.

**Sampling inspection on modules for shipment against spare parts or separate prime contract requirements shall conform to general inspection procedures of MIL-STD-105. Acceptable quality levels of 1.0% Major and 4.0% Minor shall be used for Visual and Mechanical inspection and 1.0% Major shall be used for electrical inspection.

***All electrical defects are considered major.

4.5.1.1 Order of inspection. - Group A inspection shall be performed in any order acceptable to the Government, except that the air-seal test (4.10), where applicable, shall be next to last, and the operational inspection (4.12) shall be last (see 6.6).

4.5.2 Group B inspection. - Group B inspection shall normally be performed in inspection lots that have passed Group A inspection and on samples selected from units which have been subjected to and met Group A inspection. This inspection shall conform to Table IV and to the special inspection levels of Table I of MIL-STD-105.

TABLE IV - Group B Inspection

Inspection	Req Para	Insp Para	AQL (see 4.5.2.1)
Crosstalk	3.7.2	4.8.3	4% for the group
Mark and space tone level	3.7.7	4.8.8	
Loop supply regulation	3.7.8	4.8.9	
Audio response and levels	3.7.10	4.8.12	
	3.7.11		
	3.7.20		
Automatic mark hold	3.7.22	4.8.15	
Audio squelch	3.7.23	4.8.16	
Frequency diversity	3.7.18	4.8.4	
Interchangeability	3.10	4.9	
Mechanical	3.10	4.9.2	6.5%
Electrical	3.10	4.9.1	2.5% *

4.5.2.1 Group B sampling plan. - The group B sampling plans, for the AQL's listed in Table IV shall be as follows:

AQL	Inspection level for normal and tightened inspection	Inspection level for reduced inspection
*2.5%	S-4	
4.0%	S-4	S-3
6.5%	S-4	S-3

All defects shall be considered major.

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4.5.2.2 Order of inspection within Group B. - Group B inspection shall be performed in any order which is satisfactory to the Government.

4.5.3 Group C inspection. - This inspection shall be as listed in Table V, and shall normally be performed on sample units that have been subjected to and met Group A and Group B inspection. Sample units shall be selected in accordance with 4.5.3.1.

TABLE V - Group C Inspection

Inspection	Req Para	Insp Para
(Subgroup I (see 4.5.3.1.1))		
Temperature	3.5.1	4.7.1
Low	3.5.1.1	4.7.1.1
High	3.5.1.2	4.7.1.2
BFO	3.7.9	4.8.11
Teletype Tone Reversal	3.7.5, 3.7.6	4.8.7
Scope Intensity	3.7.16	4.8.13
Transient Protection	3.7.19.1	4.8.14
Reverse Current	3.7.19.2	4.8.14
Low Voltage operation	3.7.19	4.8.18
Circuit breaker	3.7.19	4.8.19
(Subgroup II (see 4.5.3.1.2))		
Humidity	3.5.2	4.7.2
Altitude	3.5.3	4.7.3
Vibration, internal	3.5.4	4.7.4
Vibration, operating	3.5.5	4.7.5
Bounce test	3.5.6	4.7.6
Shock test; bench handling	3.5.6	4.7.7
Immersion	3.5.7	4.7.8
720 Hours run	3.7.28	4.8.20
Teletype tone stability	3.7.3	4.8.5
Error rate vs Signal-to-noise ratio	3.7.21	4.8.2
Fungus	3.4.5.2	4.7.9

4.5.3.1 Sampling for Group C inspection. - Units selected for each Group C inspection shall be selected without regard to their quality in accordance with the following:

4.5.3.1.1 Subgroup I. - For this subgroup, two units from the first 50 production units shall be selected for the first Group C inspection. For subsequent Group C inspection, two units per month shall be selected.

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4.5.3.1.2 Subgroup II. - For this subgroup, two units from the first 50 production units shall be selected for the first Group C inspection. For subsequent Group C inspections, one unit from each 250 (or fraction thereof) production units shall be selected.

4.5.3.2 Order of inspection. - Group C inspection shall be performed in any order acceptable to the Government.

4.5.3.3 Noncompliance. - The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process, and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Groups A and B inspection may be reinstituted; however, final acceptance will be withheld until the Group C reinspection results have shown that the corrective action was effective (see 6.5).

4.5.4 Reinspection of conforming Group B and Group C sample units. - Unless otherwise specified, sample units which have been subjected to and passed Group B and Group C inspection or both, may be accepted on contract, provided that they are resubjected to and pass Group A reinspection after repair of all visible and known damage.

4.6 Preconditioning. - The tests of 4.6.1 and 4.6.2 shall be performed on all equipments on order. Bounce preconditioning shall precede Burn-in/Debugging.

4.6.1 Bounce preconditioning (see 3.11.1). - The unit, with shock mounts (if any) removed or blocked, shall be placed in its normal operation position on the table of the Type 1000-SC Package Tester as made by the L. A. B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for one (1) minute. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.6.2 Burn-In/Debugging. -
4.6.2.1 Sequence of testing. - Before subjecting equipments to this procedure, the equipment must have successfully met the requirements of paragraph 3.7.4 after the test of paragraph 4.8.6.

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4.6.2.2 Test Conditions. - The equipment shall be operated for a failure free period of 36 hours when subjected to the conditions of Test Level B of MIL-STD-781. The System Test 4.8.2 shall be performed after burn-in/debugging.

4.6.2.3 Failures. - If the equipment fails the system test during or at the end of burn-in/debugging period, the failure cause shall be corrected and the equipment re-subjected to the 36 hour burn-in/debugging period. If an equipment fails Group A, Group B or Group C tests, the failure cause shall be corrected and the equipment re-subjected to the 36 hour burn-in/debugging.

4.6.2.4 Compliance. - Units passing burn-in/debugging will be submitted to Group A, Group B and Group C inspection in that order without benefit of repair, cleaning, realignment or other changes unless otherwise specified herein.

4.7 Service conditions tests (see 3.5).

4.7.1 Temperature (see 3.5.1).

4.7.1.1 Low temperature. - The equipment shall be tested in accordance with Method 502, Procedure 1, of MIL-STD-810. The storage temperature shall be -80°F for 24 hours and the operating temperature shall be -40°F. The equipment shall be operated until temperature stabilization is reached. Prior to testing the equipment shall meet full specification performance for the measurements of 3.5.1.1.

4.7.1.2 High temperature. - The equipment shall be tested in accordance with Method 501, Procedure I, of MIL-STD-810. The operating temperature shall be +150°F. The equipment shall be operated until temperature stabilization is reached. Prior to testing, the equipment shall meet full specification performance for the measurements of 3.5.1.2.

4.7.1.3 The above tests shall be performed with an input voltage of 27.0 VDC. The tests shall be repeated with an input voltage of 24.0 VDC and 30.0 VDC. The tests involving tone stability (3.7.3, 4.8.5) shall be made within the first 10 minutes after the modem is turned "ON". The BFO stability requirements (3.7.9, 4.8.11) shall be met within the first 15 minutes of operation.

4.7.2 Humidity (moisture-resistance) (see 3.5.2). -

4.7.2.1 Test conditions. - The equipment shall be tested in accordance with Method 507, Procedure II of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.5.2. Measurements shall be made during the periods shown on the figure 507-2 for the cycle. During the fifth cycle, the equipment shall be removed from its case. Do not remove equipment from the humidity chamber for measurements, and do not leave power on after measurements have been completed. If any equipment fails to meet the specified performance (3.5.2)

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during cycling, it shall be realigned or readjusted once and the test continued. No repair or replacement of parts shall be permitted. If the equipment still fails to meet the specified performance (3.5.2) during cycling, or fails subsequently, it does not pass the test. In addition, if the equipment fails to meet specified requirements after final conditioning and adjustment, it does not pass the test.

4.7.3 Altitude (3.5.3). - The equipment shall be tested in accordance with Method 500, Procedure 1, of MIL-STD-810. Prior to, during and after testing, the equipment shall meet full specification performance for the measurements specified in 3.5.3.

4.7.4 Test for internal vibration. - Internal vibration of the equipment shall be measured as follows, to determine conformance to 3.5.4:

a. Secure the equipment (shock mounts other than those on internal components, removed or blocked) directly to a vibration table that can be controlled to within 10 percent of the specified amplitude. Attachment points shall simulate chassis securing points. Mounting method shall be such that vibration within the equipment can be observed and measured. To facilitate this observation and measurements, subassemblies may be tested separately provided that they are secured to the table in a manner similar to that used to mount them in the equipment.

b. Vibrate the equipment successively in the three mutually perpendicular planes over a frequency range of 10 to 55 Hz. The total excursion of the applied vibration shall not be less than 0.030 inch.

c. In each of the three directions, change the frequency in steps of one Hz and maintain each frequency for at least 10 seconds.

d. Measure the vibration amplitudes by optical means or by other means, provided that vibration of the part is not affected by the measurement.

4.7.5 Test for equipment vibration. - The modem, in its case, shall be tested as follows to determine conformance to 3.5.5:

a. Secure the equipment on Mounting MT-3140( )/GRC-106 less shock isolators, to a vibration table that can be controlled within 10 percent of the specified amplitude. The equipment shall be mounted in its normal operating position with connections (such as cable assemblies) in place.

b. Vibrate the complete equipment assembly, successively, in three mutually perpendicular directions, over a frequency range of 10 to 55 Hz. The total excursion of the applied vibration shall be 0.040 inch in the horizontal plane and 0.060 inch in the vertical plane.

c. Vibrate the equipment for 30 minutes in each of three directions for a total test time of 90 minutes. Change the vibration frequency during the test at the rate of 1 Hz.

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4.7.6 Bounce (see 3.5.6). - The equipment shall be tested in accordance with Method 514, Procedure IX, Part 2, of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.5.8. The modem shall be secured to the 48 inch by 48 inch vehicular adapter plate using Mounting MT-3140( )/GRC-106 and shock isolators. Before, during and after each 45 minute phase of step 3, the equipment shall be subjected to the tests of 3.5.6 except that during the bounce test, perform the tests of 3.5.6 in the 85 Hz mode only.

4.7.7 Shock test; bench-handling (3.5.6). - The equipment shall be tested in accordance with Method 516, Procedure V of MIL-STD-810. Prior to and after testing, the equipment shall meet full specification performance for the measurements specified in 3.5.6.

4.7.8 Immersion (see 3.5.7). - The equipment, shall be immersed to a minimum depth of 3 feet of fresh water for two hours. Immediately prior to immersion, the temperature of the equipment shall be 40°F, or more, above the temperature of the water. The tank in which the equipment is immersed shall be of sufficient capacity to maintain the water within ± 2°F of its initial temperature or the temperature of the water shall be maintained within those limits by other means. After completion of the 2-hour period of immersion, the equipment shall be removed from the water and wiped dry on exterior surfaces. When the equipment is opened, there shall be no evidence of leakage. The three feet shall be measured from the surface of the water to the topmost point on the equipment. Breather valve shall be plugged up during this test.

4.7.9 Fungus test (see 3.4.5.2). - The equipment (including accessories) shall be tested in accordance with Method 508, Procedure I of MIL-STD-810. Spraying shall include parts and materials on and under the chassis, and the interior of the equipment case. The equipment shall remain in an open condition during the test. Operation is not required.

4.7.10 Inspection of Integrated circuit (IC's) (see 3.4.6). - Quality conformance inspection for IC's shall be in accordance with MIL-S-19500. First Article testing in accordance with general requirements of SC-A-46600 shall be performed in lieu of the qualification testing requirements of MIL-S-19500.

4.8 Electrical Tests (see 3.7).

4.8.1 Standard test conditions. - Unless otherwise specified, the following standard test conditions shall apply:

4.8.1.1 Test conditions. - All tests shall be made in accordance with test conditions (standard ambients) Paragraph 3, General Requirements of MIL-STD-810.

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4.8.1.2 27.0 VDC, Power source. - The power source shall be 27.0 ± .5 volts, direct current (vdc), except where specifically changed in the detailed procedure. The regulation shall be such as to maintain this voltage when the load is changed from no load to the maximum for any of the equipments under test. Noise and ripple voltage of the source shall be less than 0.5 volts, ac. The voltage source shall be adjustable over the range of 22 to 30 volts.

4.8.1.3 Grounding. - A single ground connection shall be provided which shall connect to the internal grounding system of the equipment. Test equipment shall be grounded to the common equipment ground whenever practicable. The cases of metal cased test equipment shall be grounded and so noted in test record.

4.8.1.4 Duty cycle. - The duty cycle should be continuous unless otherwise specified.

4.8.1.5 Warm-Up Period. - Unless otherwise specified, measurements shall be made after five (5) minutes from the time primary power is applied.

4.8.2 System Test. - The system test shall be performed using the systems test set SM-D-583659 and error detector and counter SM-D-583656. Test fixtures shall be furnished by the contractor. During the system test, the following shall be recorded:

a. Error rate vs signal-to-noise ratio. Enough data will be recorded to present a smooth curve from 10,000 errors in 100,000 bits to 1.0 error in 100,000 bits (see 3.7.21).

b. Average bias distortion in Receive loop. This measurement shall be made using an oscilloscope or VTVM across loop load resistance (see 3.7.13).

c. Instantaneous distortion in Receive loop. This measurement shall be made using a suitable oscilloscope (see 3.7.13).

d. Teletype tone stability - This measurement shall be made periodically during the system test when the modem shall be operated in steadyMark or Space (3.7.3).

e. Teletype tone (Mark and Space) level (see 3.7.7).

The above data shall be recorded for all teletype modes in the modem and at both 60 and 100 wpm. The system test shall be performed as part of the following tests in 60 wpm 85 Hz and 850 Hz modes except that error rate will be measured with a fixed noise level at 100 wpm. The noise level shall be chosen so that the error rate is within 20 to 500 per 100,000 bits. During these tests, an increase in the error rate to more than 5 times the chosen error rate shall be cause for rejection.

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(1)	Temperature Test	4.7.1
	Low	4.7.1.1
	High	4.7.1.2
(2)	Humidity (Moisture-resistance)	4.7.2
(3)	Bounce Test	4.7.6
(4)	Vibration Test	4.7.5
(5)	Altitude (elevation)	4.7.3
(6)	Shock, bench handling	4.7.7
(7)	720 Hour test	4.8.20

4.8.3 Crosstalk (see 3.7.2). - The modem shall be connected to the test fixture and set up for operation on 85-Hz plus voice mode. An audio oscillator shall be connected to the audio input of the modem and a wave analyzer connected to the audio output. Voice will be simulated by using the audio oscillator at 1,000 Hz. With the audio levels adjusted properly, measure a steady teletype tone relative to the 1-KHz audio tone. The TTY tone must be at least minus 46 db relative to the 1-KHz audio tone. Repeat using audio oscillator frequencies of 350 Hz and 2,200 Hz. Measure the 1-KHz tone relative to the TTY tone at TP1-A1 on the receiver module. The 1-KHz tone should be minus 46 db relative to the TTY tone.

4.8.4 Frequency diversity (see 3.7.18). - The modem shall be connected to the diversity test fixture SM-D-145895. The test fixture shall incorporate two audio filters, one high pass and one low pass with a cross over of approximately 1000 Hz. An attenuator shall be used at the output of each filter. With the modem in 85 Hz operation, a noise source shall be added to provide between 20 and 500 errors in 10^5 bits. With the modem in the 85 Hz diversity mode and with equal channel levels, the resulting errors shall be within the 85 Hz Diversity Curve of Figure 1. Attenuate each filter output alternately by 40 db. The receive signal shall continue with an error rate within the 85 Hz Diversity specification.

4.8.5 Teletype tone stability (see 3.7.3). - The stability of the MARK and SPACE tones on 85 Hz NSK, 850 Hz FSK and 85 Hz diversity shall be measured as part of the system test of 4.8.2. Data pertaining to tone stability shall be recorded as part of the system test and shall also be recorded separately to determine the trend in the long term stability of the TTY tones. During temperature test, tone stability shall be measured until the equipment reaches thermal equilibrium. The equipment shall conform to the requirement of 3.7.3.

4.8.6 Teletype tone frequency adjustment (see 3.7.4). - The teletype tones shall be adjusted to show conformance with the requirements of 3.7.4. The tones shall then be adjusted as accurately as possible to the correct frequencies. This test shall be performed just prior to the start of the burn-in/debugging test of 4.6.2. There shall be no further adjustment in tone frequency so that the data on tone stability will be valid.

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4.8.7 Transmit (Teletype) tone reversal (see 3.7.5) and Receive polarity Reversal (see 3.7.6). - Connect the equipment to the test fixture. Apply a steady MARK signal to the modem. Attach a DC current meter in series with a resistive load to the output loop of the modem. Observe a steady current through the meter. Switch the teletype tone reversal switch on the transmit module. The meter should read ZERO current. Switch the receive polarity reversal switch behind the access door on the front panel. The meter should return to the steady current condition.

4.8.8 Mark and Space Tone Level (see 3.7.7). - Attach a wave analyzer to the transmit audio output. Measure the MARK and SPACE tone level. Adjust teletype tone levels in the 85 Hz mode, 850-Hz mode and diversity mode. The relative level of a tone pair shall be maintained within 2.0 db during the temperature test of 4.7.1.

4.8.9 Loop supply regulation (see 3.7.8). - Connect a DC current meter and variable resistor to the "Send" loop. Set the loop current switch to "20 ma". Vary the resistor from zero ohms to 1300 ohms. Repeat the test with the loop current switch set at "60 ma". Repeat entire test on "Receive" loop. The result shall meet the requirements of 3.7.8.

4.8.10 BFO range (see 3.7.9). - Connect a counter to the BFO output. After a five minute warm-up, measure the frequency of the BFO at the extremes of the BFO control. The frequency spread shall meet the requirements of 3.7.9.

4.8.11 BFO stability (see 3.7.9). - During the temperature test of 4.7.1 the frequency of the BFO shall be checked. The test shall be continued until the equipment reaches thermal equilibrium without adjustment of the BFO control. The frequency of the BFO shall meet the requirements of 3.7.9.

4.8.12 Audio response and levels (see 3.7.10, 3.7.11 and 3.7.20). - The modem shall be connected to the test fixture and an audio oscillator connected to the audio input. A counter and VTVM shall be connected to the audio output. The frequency response of the system shall be measured in both the "Voice only" and "85 Hz plus Voice" modes. The response shall meet the requirements of 3.7.20. During this test, the audio output level shall be measured at both the 10 milliwatt output and the 2 watt output. Distortion shall also be measured. The requirements of 3.7.11 shall be met. The variation levels of voice and teletype tones in the 85 Hz plus voice mode shall be measured. Optimum ratio between voice and teletype tones shall be established per paragraph 2.7 and 2.11 of ECOM Drawing SM-A-583645.

4.8.13 Scope Intensity (see 3.7.16). - The modem shall be operated in each of its modes. With no signal input, the scope intensity shall be squelched so that no trace or dot shall appear. A signal shall be applied and the scope pattern checked for brightness. The requirements of 3.7.16 shall be met.

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4.8.14 Transient protection (see 3.7.19.1). - The voltage transients shall be applied to the input power line. The transients shall vary in voltage up to 100v peak and the duration and repetition rate shall be varied from 10 to 100 Ms and from .5 to 5 pulses per second respectively. The modem shall be checked for compliance with 3.7.19.1. During this test, the DC input voltage shall be reversed and the equipment checked for compliance with 3.7.19.2.

4.8.15 Automatic mark hold (see 3.7.22). - The modem shall be connected to the test fixture and a noise source attached. The modem shall be operated with varying signal-to-noise ratio and the signal removed at each chosen S/N ratio. The operation of the automatic Mark hold shall be checked for compliance with 3.7.22.

4.8.16 Audio squelch (see 3.7.23). - The modem shall be connected to the test fixture and a source of noise added. The modem shall be modulated by a voice message and the squelch operation checked with varying signal to noise ratios for compliance with 3.7.23.

4.8.17 Front panel controls and receptacles. - Test shall be performed to demonstrate the functioning of the following controls or receptacles for compliance with the requirements:

Aux Receptacle - 3.7.25 and 3.7.26
Ground Post - 3.7.29

4.8.18 Low voltage operation (see 3.7.19). - The modem shall be connected to the test fixture and the system test, 4.8.2, performed with input power line voltage of 22.0 VDC. The requirements of 3.7.19 shall be met.

4.8.19 Circuit breaker test (see 3.7.19). - The circuit breaker shall be checked for its overload opening point. The requirements of 3.7.19 shall be met.

NOTE: If circuit breaker does not immediately trip, switch the power supply to "Off" in order to prevent damage to the modem.

4.8.20 720 Hour test (see 3.7.28). - The modem shall be operated continuously on a system basis, using reversals, for a period of 720 hours. At the start of this test and during the test, the system test (4.8.2) shall be performed daily and the results recorded. At the end of the 720 hour run, the system test shall be performed and any degradation noted (see 3.7.28).

NOTE: During daily performance measurements on the modem, data pertaining to tone stability shall be recorded separately to show the trend in long term stability of the TTY tones. At the end of the 720 hour run, the equipment shall conform to the requirement of 3.7.3.

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4.8.21 One Way/Duplex Switch (see 3.7.25). - Position One Way/Duplex switch to Duplex and observe with ohmmeter that J5 terminal "A" is approximately 14.5 ohms to ground. The requirements of 3.7.25 shall be met.

4.8.22 Send/Receive Switch (see 3.7.26)

a. Position One Way/Duplex switch of test set to one way. Position modem One Way/Duplex switch to one way. Observe that Send/Receive switch keys modem.

b. Observe that in Rec position there is continuity between pins D and E of J5 and in send position Pin D of J5 has continuity to ground.

c. The requirements of 3.7.26 shall be met.

4.8.23 Audio Functions - Mode selector switch (see 3.7.27). - In the teletype only modes of operation, the teletype tones shall be at the audio outputs, Pin A of J3, Pin A of J4 and Pin L of J4. In the 85 Hz + Voice mode, only the voice portion of the signal shall appear at the above outputs. In the Voice only mode, the audio output shall appear at the above outputs. The requirements of 3.7.27 shall be met.

4.8.24 Test Meter (see 3.7.17). - The meter functions will be checked at each position of the meter switch:

a. Postion Test Meter function switch on modem to DC Loop No. 1.

b. Connect milliammeter in series with Loop inputs and observe that meter is within 10% accuracy for Loop 1 and 2 Positions.

c. Position Test Meter Function switch to Discriminator position and observe left swing for Mark and right swing for Space approximately within meter marking.

d. Position Test Meter Function switch to regulated 20-V position and observe that the pointer is within 10% of arrow mark.

e. Position Test Meter Function switch to Receive position. Position Gain control of Test Set to 2.45 V (10 MW) and observe that Test Meter Pointer is within the Receive range.

f. The requirements of 3.7.17 shall be met.

4.9 Inspection for Interchangeability. - The electrical performance listed in 4.9.1 shall be measured and the mechanical dimensions listed in 4.9.2 shall be gaged to determine conformance to 3.10. When a performance requirement or listed dimension is not within specified or design limits, it shall be considered a major defect.

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4.9.1 Electrical performance. - Prior to assembly into the equipment, the plug-in modules and subassemblies listed below shall be tested to determine compliance with the performance requirements specified in the applicable test procedure.

Front Panel and Chassis
Receiver Audio and BFO
Scope
Transmitter
Receiver
Loop Battery

4.9.2 Inspection for Dimensional Interchangeability. - The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.10. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers and mountings.

d. Location of connectors, locking pins, fasteners, slides and mountings which receive mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

4.10 Air seal test. - The equipment shall be opened and closed again in such manner as to break and remake the seal. Immediately thereafter, the equipment as field transported, shall be subjected to a vacuum of 1 pound per square inch (1 pound per square inch less than the atmospheric pressure surrounding the equipment) applied to the interior of the transit case or to the interior of the equipment inclosure when no transit case is provided. The vacuum then shall be valved-off and the interior pressure measured during the ensuing period of 1 minute. The gage used for measurement of the vacuum shall be of such accuracy that a difference of .01 psig can be determined readily. The requirement of 3.12 shall be met.

4.11 Visual and Mechanical inspection. - Visual and mechanical inspection shall be in accordance with MIL-STD-252 and the following inspection: Parts, components and complete equipments shall be inspected for workmanship, mechanical construction, loose nuts and screws, application of specified treatments and miscellaneous defects. The equipment and controls shall be inspected for proper mechanical and electrical operation. Wiring, soldered connections, grounds, welds, finishes, etc. shall be inspected for workmanship. Clearances,

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dimensions, and mechanical adjustments shall be measured. Labels and markings shall be inspected for conformance with approved facsimiles, when facsimiles are required. Examination also may be made for other visual or mechanical defects, similar to those described above, that are contrary to specified requirements for the equipment. Visual and mechanical defects shall be classified by the Government as major or minor in accordance with the definitions of MIL-STD-105.

4.12 Final function check. - Immediately prior to packaging the equipment for shipment, the contractor shall perform a check on all units produced to determine proper functioning of all controls, indicators and modes of operation.

4.13 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of section 5.

4.14 Rough handling test (preparation for delivery). - When a rough handling test in accordance with Specification MIL-P-116 is required by the contract (see 6.2 (i)), the final functional tests (4.12) shall be conducted to determine freedom from operational malfunction caused by rough handling.

5. PREPARATION FOR DELIVERY.

5.1 Preservation and packaging. - Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A:

5.1.1.1 Cleaning. - Modem, Radio Teletypewriter Set MD-522( )/GRC shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. - Modem, Radio Teletypewriter Set MD-522( )/GRC, shall be dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.3 Preservation application. - None required.

5.1.1.4 Unit packaging. - Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. - Each technical literature shall be packaged method 1C-1.

5.1.1.4.2 Modem, Radio Teletypewriter Set MD-522( )/GRC. - Each Modem, Radio Teletypewriter Set MD-522( )/GRC, shall be individually packaged method III as follows:

Cushion each item on all surfaces with cells or pads or both, fabricated of fiberboard conforming to PPP-F-320, type CF, class weather-resistant, variety SW, grade W5c, designed to protect all projections and absorb the shock

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of impact in handling and transit. Place the cushioned unit within a close-fitting fiberboard box, conforming to PPP-B-636, W5c. Place the technical literature, packaged as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level C. - Modem, Radio Teletypewriter Set MD-522( )/GRC, shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing. - Packing shall be level A, B, or C, as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents (see 6.2b).

5.2.1 Level A.

5.2.1.1 Palletized load. - A quantity of Modem, Radio Teletypewriter Set MD-522( )/GRC, packaged as specified in 5.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147, except that the pallet shall be soft wood conforming to NN-P-71, type IV, size 2. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping.

5.2.1.2 Less than palletized load. - When quantities per destination are less than a pallet load, the item packaged as specified in 5.1, shall be waterproofed, with tape conforming to PPP-T-76, in accordance with taping requirements of the appendix of the box specification. A quantity of the waterproofed containers shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, class B.

5.2.2 Level B.

5.2.2.1 Palletized load. - A quantity of Modem, Radio Teletypewriter Set MD-522( )/GRC, packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1.

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5.2.2.2 Less than palletized load. - When quantities per destination are less than a pallet load, a quantity of Modem, Radio Teletypewriter Set MD-522( )/GRC, packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant, variety DW. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, type IV. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97, type IV and PP-S-760, type II, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.3 Level C.

5.2.3.1 Palletized load. - A quantity of Modem, Radio Teletypewriter Set MD-522( )/GRC, packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1 except that the fiberboard caps shall be class domestic.

5.2.3.2 Less than palletized load. - When quantities per destination are less than a pallet load, a quantity of Modem, Radio Teletypewriter Set MD-522( )/GRC, packaged as specified in 5.1, shall be packed as specified in 5.2.2.2 except that the fiberboard boxes shall conform to PPP-B-640 and PPP-B-636, class 1 and class domestic, respectively, and reinforcing shall not be required for boxes conforming to PPP-B-636.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES.

6.1 Intended Use. - Modem, Radio Teletypewriter Set MD-522( )/GRC is intended to provide an FSK facility for use with the HF, SSB family of radio sets.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Levels of preservation, packaging, and packing required (Level A, level B, or level C) (see Section 5).

c. Ancillary equipment to be procured (see 3.9).

d. First Article inspection:

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(1) Four sample units of each item cited in section 1 are generally required so that lengthy environmental and electromagnetic compatibility tests can be completed on two sample units while complete performance measurements can be made on the other sample units (see 3.2).

(2) Detailed requirements covering First Article pack(s), including inspection to be performed thereon and descriptive details, plans, drawings, etc. (NOTE: These may be obtained upon request from Packaging Branch, Engineering Support Division, US Army Electronics Command, Philadelphia, Pennsylvania 19103).

e. Marking and shipping of samples.

f. Place of final inspection.

g. Technical literature required (see 3.14).

h. Quantity of tools and running spare parts required (see 3.14).

i. When the rough handling test and functional test are required (see 4.14).

6.3 Color (see 3.4.1) - The color chip furnished by the procuring agency will be color chip No. X-24087, and may be obtained upon request to Commander, US Army Electronics Command, ATTN: AMSEL-PP, 225 South Eighteenth Street, Philadelphia, Pennsylvania 19103.

6.4 Definitions of Level B. - Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments; covered storage for a period of three years or less; and distribution within the continental limits of the United States.

6.5 Group C Inspection. - Approval to ship may be withheld, at the discretion of the Government, pending the decision from the Contracting Officer on the adequacy of corrective action (see 4.5.3.3).

6.6 Location of air seal inspection. - It is desirable that the air seal test (4.10) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. Any preparation for shipment which would require breaking of the equipment seal should be accomplished prior to the air seal test so that the seal may remain intact thereafter. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

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6.7 Nomenclature. - The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example, MD-522W/GRC. The contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.1).

6.8 Verification inspection (Government) (see 4.5). - The extent of Government verification inspection shall be determined by the Government and will include maximum utilization of the contractor's inspection records and the quality history of the product as reflected in these records. Such inspections will usually not exceed the definitive quality assurance provisions established in the specification, contract or order; however, the Government reserves the right to perform any amount of inspection (including 100% inspection) when the Contracting Officer determines such additional inspection is required.

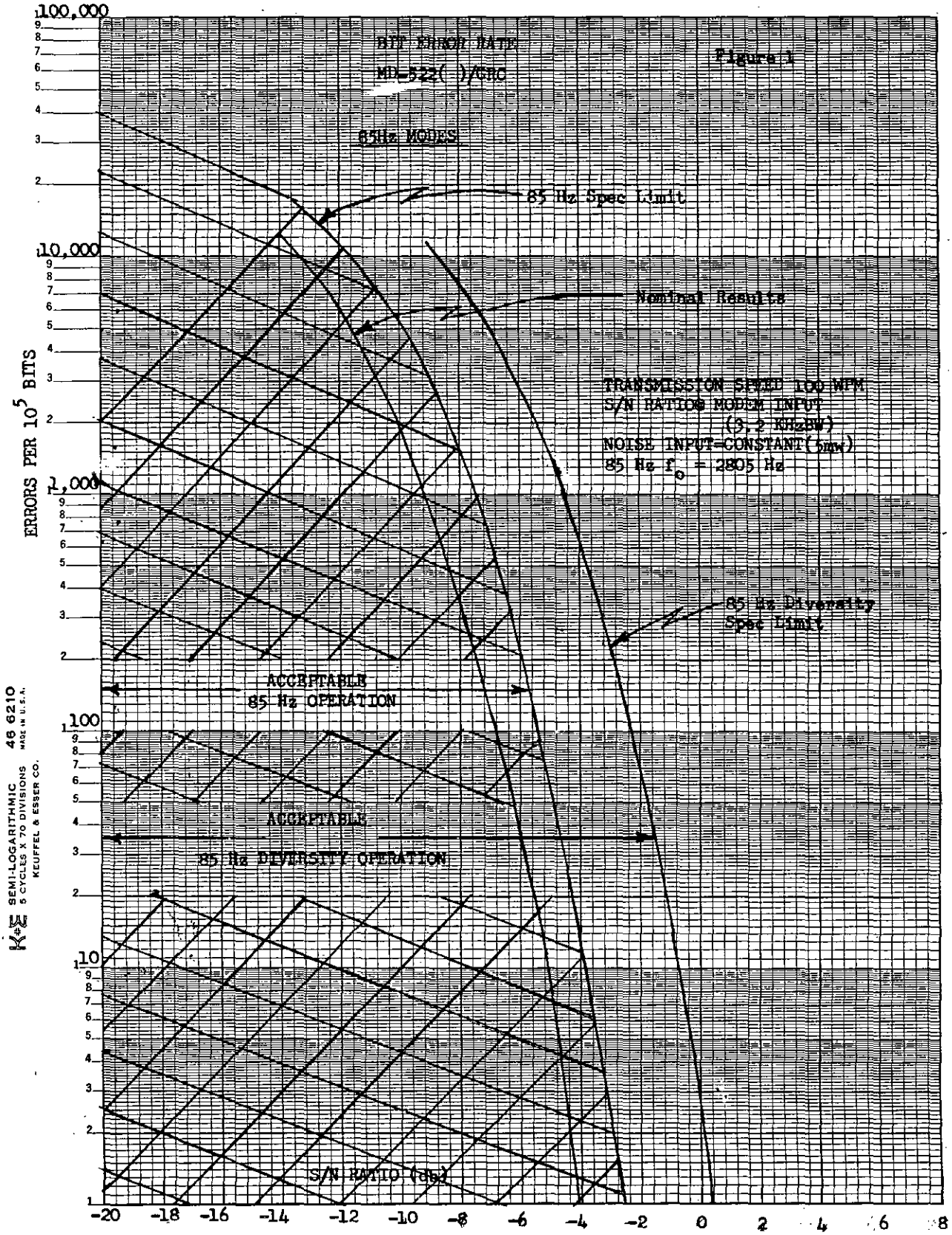
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Preparing Activity
Army-EL

Project Number 5815-A087

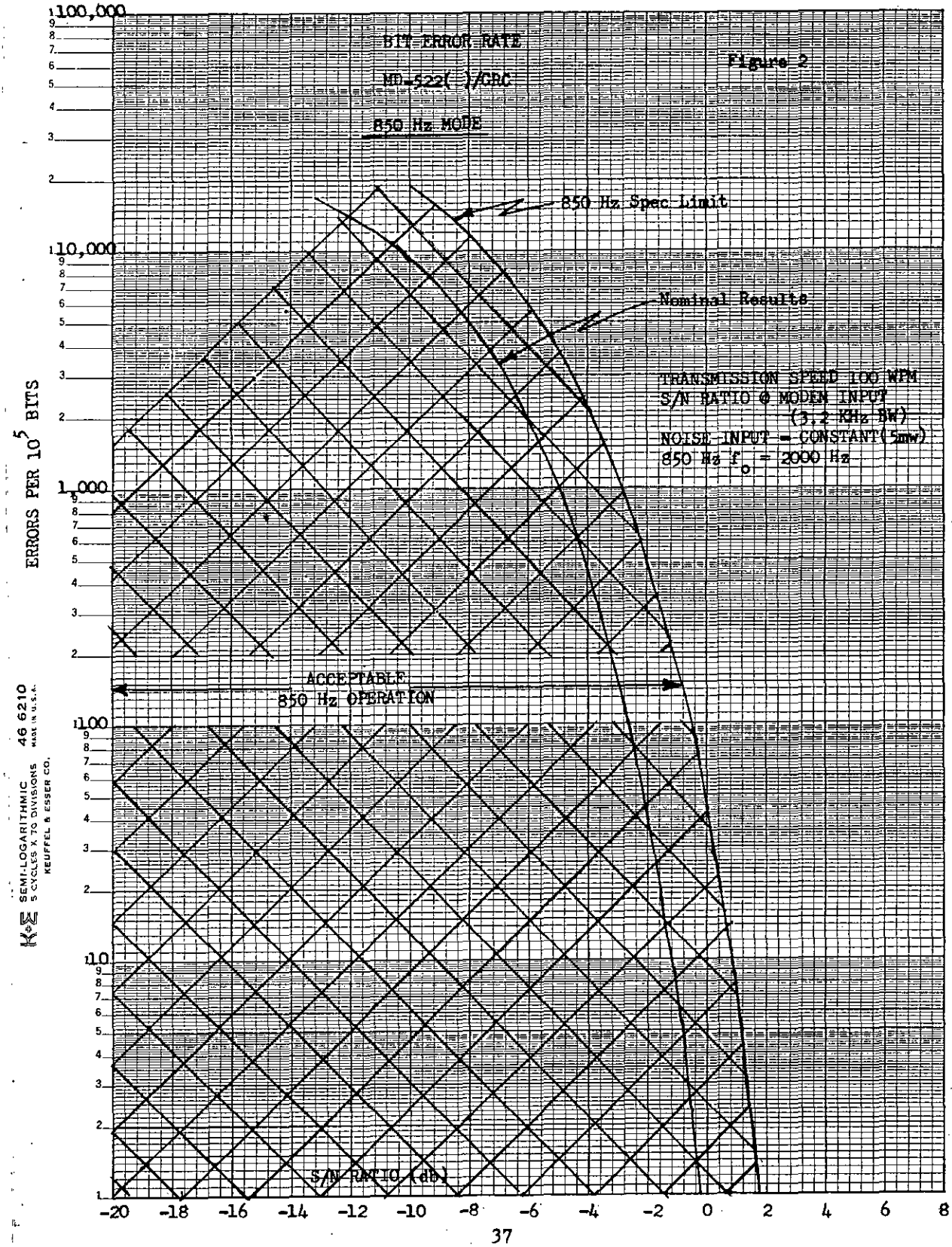
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Figure 1



K&E SEMI-LOGARITHMIC 46 6210
5 CYCLES X 70 DIVISIONS MADE IN U.S.A.
KEUFFEL & ESSER CO.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-M-55529A(EL) MODEM, RADIO TELETYPEWRITER SET MD-522( )/GRC

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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